1

We know that all the components of $\vec{E}$ and $\vec{B}$ are packed into the tensor $F_{\mu\nu}$, and we know how to transform the components of a tensor to a different reference frame using the Lorentz transform:

$$F_{\mu'\nu'} = \Lambda^{\mu}{}_{\mu'}\Lambda^{\nu}{}_{\nu'}F_{\mu\nu}, \tag{5}$$

where the Lorentz transform matrix along the $+x$ direction (technically the inverse Lorentz transform matrix since we are transforming a (0,2) tensor, similar to how we transform a one-form) is given by

$$\Lambda^{\mu}{}_{\mu'} = \begin{bmatrix} \gamma & \gamma v & 0 & 0 \\ \gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

The safest way to compute Eq. (5) is simply to sum up μ and ν. For example, to find out E_1', we can compute

$$\begin{aligned} F_{0'1'} &= \Lambda^{0}{}_{0'}\Lambda^{1}{}_{1'}F_{01} + \Lambda^{1}{}_{0'}\Lambda^{0}{}_{1'}F_{10} \qquad \text{(I have omitted terms that will result in 0)} \\ &= -\gamma^2 E_1 + \gamma^2 v^2 E_1 = -\gamma^2(1-v^2)E_1 = -E_1. \end{aligned} \tag{7}$$

Then $E_1' = -F_{0'1'} = E_1$.

Similarly, to find out E_2', we can compute

$$\begin{aligned} F_{0'2'} &= \Lambda^{0}{}_{0'}\Lambda^{2}{}_{2'}F_{02} + \Lambda^{1}{}_{0'}\Lambda^{2}{}_{2'}F_{12} \\ &= -\gamma E_2 + \gamma v B_3 = -\gamma(E_2 - vB_3), \end{aligned} \tag{8}$$

so we find $E_2' = -F_{0'2'} = \gamma(E_2 - vB_3)$.

Performing similar calculations for the rest, we find

$$\begin{aligned} E_1' &= E_1 \\ E_2' &= \gamma(E_2 - vB_3) \\ E_3' &= \gamma(E_3 + vB_2), \end{aligned} \tag{9}$$

and

$$\begin{aligned} B_1' &= B_1 \\ B_2' &= \gamma(vE_3 + B_2) \\ B_3' &= \gamma(B_3 - vE_2). \end{aligned} \tag{10}$$

Note that we can also use matrix multiplications to compute Eq. (5), but we must be careful not to apply matrix multiplications blindly. Recall that $F_{\mu\nu}$ does not behave like a matrix in the language of linear algebra. However, if we arrange the elements of the inverse Lorentz transform and the tensor $F_{\mu\nu}$ into matrices in a careful way, the matrix multiplication can still give us the correct answer. Note that Eq. (5) can be written as

$$F_{\mu'\nu'} = \Lambda^{\mu}{}_{\mu'}F_{\mu\nu}\Lambda^{\nu}{}_{\nu'}. \tag{11}$$

By matrix algebra, we know that if a matrix A is the product of matrices B, C, and D, then A's elements can be written as

$$(A)_{\mu'\nu'} = (B)_{\mu'\mu}(C)_{\mu\nu}(D)_{\nu\nu'}. \tag{12}$$

Comparing this with Eq. (11), we conclude that

$$(F')_{\mu'\nu'} = (\Lambda^T F \Lambda)_{\mu'\nu'}, \tag{13}$$

where we have arranged the elements of the inverse Lorentz transform and the tensor $F_{\mu\nu}$ into matrices such that $(F)_{\mu\nu} = F_{\mu\nu}$, $(\Lambda)_{\mu\mu'} = \Lambda^{\mu}{}_{\mu'}$, and $(F')_{\mu'\nu'} = F_{\mu'\nu'}$. Again, when we write a tensor as a matrix, the matrix does not necessarily have the same physical meaning of the tensor. The matrix is just a tool for us to display the components of the tensor and carry out calculations, and we must be careful about how the calculation is performed to ensure the matrix multiplication gives a result that is consistent with tensor operations.

2

Recall the Levi-Civita symbol gives 1 if the indices are even permutation of $\{0, 1, 2, 3\}$ and gives -1 if the indices are odd permutations of $\{0, 1, 2, 3\}$ and is 0 if there are repeated indices. Then the computation is straightforwad, and we get

$$*F^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\lambda\rho}F_{\lambda\rho} = \begin{bmatrix} 0 & B_1 & B_2 & B_3 \\ -B_1 & 0 & -E_3 & E_2 \\ -B_2 & E_3 & 0 & -E_1 \\ -B_3 & -E_2 & E_1 & 0 \end{bmatrix}. \tag{14}$$

3

From the diagram, we see that all four worldlines start from $(t = 1s, y = 5m)$ and end at $(t = 3s, y = 5m)$. If there is no gravity, we have seen in lecture (this was discussed in section 6.1 and 6.2) that among all observers who are traveling through spacetime between two events, the inertial observe experiences the maximum amount of proper time. The problem has gravity, but by the Equivalence Principle, an observer who is freely-falling in a gravitational field feels just like an inertial observer, given that only local measurements are made. Therefore, in a gravitational field, the freely-falling observer experiences the maximum proper time. We see that curve C resembles the worldline of a freely-falling object, so C has the greatest proper time.

4

From the lecture notes, we have seen that the ratio of proper times between a clock located at A and a clock located at B on the earth (with different altitudes) is given by

$$\frac{\Delta\tau_A}{\Delta\tau_B} = Exp\left[\frac{Gm_E}{c^2}\left(\frac{1}{r_B} - \frac{1}{r_A}\right)\right]. \tag{15}$$

Assuming D.C. is located at 0 altitude, we take $r_{DC} = r_{earth} = 6378000m$. Denver is $1600m$ above D.C., so $r_{Denver} = 6379600m$. Plugging these into the equation above, we get

$$\frac{\Delta\tau_{Denver}}{\Delta\tau_{DC}} = Exp\left[\frac{Gm_E}{c^2}\left(\frac{1}{r_{DC}} - \frac{1}{r_{Denver}}\right)\right] = Exp(2\times 10^{-13}) = 1 + 2\times 10^{-13}. \tag{16}$$

This means that a clock in Denver runs faster by 2×10^{-13} s for every second passed in D.C.

5

(a) The original geodesic equation is

$$\frac{d^2x^\beta}{d\tau^2} + \Gamma^\beta{}_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0. \tag{17}$$

Now we want to parametrize the geodesic with a different parameter σ. By the chain rule, we know that

$$\frac{dx^\beta}{d\tau} = \frac{dx^\beta}{d\sigma}\frac{d\sigma}{d\tau}. \tag{18}$$

Applying Eq. (18) to Eq. (17), we get

$$\frac{d}{d\tau}\left(\frac{dx^\beta}{d\sigma}\frac{d\sigma}{d\tau}\right) + \Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{d\sigma}{d\tau}\frac{dx^\nu}{d\sigma}\frac{d\sigma}{d\tau} = 0 \tag{19}$$

$$\frac{d^2x^\beta}{d\sigma^2}\left(\frac{d\sigma}{d\tau}\right)^2 + \frac{dx^\beta}{d\sigma}\frac{d^2\sigma}{d\tau^2} + \Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}\left(\frac{d\sigma}{d\tau}\right)^2 = 0 \tag{20}$$

$$\left(\frac{d^2x^\beta}{d\sigma^2} + \Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}\right)\left(\frac{d\sigma}{d\tau}\right)^2 = -\frac{dx^\beta}{d\sigma}\frac{d^2\sigma}{d\tau^2} \tag{21}$$

$$\frac{d^2x^\beta}{d\sigma^2} + \Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} = -\left(\frac{d\sigma}{d\tau}\right)^{-2}\frac{d^2\sigma}{d\tau^2}\frac{dx^\beta}{d\sigma}. \tag{22}$$

Now let's examine the right-hand side of Eq. (22), specifically the term $-\left(\frac{d\sigma}{d\tau}\right)^{-2}\frac{d^2\sigma}{d\tau^2}$. We set $u = \frac{d\sigma}{d\tau}$, then the term becomes

$$-u^{-2}\frac{du}{d\tau} = -\frac{1}{u}\left(\frac{1}{u}\frac{du}{d\tau}\right) = -\frac{1}{u}\frac{d}{d\tau}\ln(u) = \frac{1}{u}\frac{d}{d\tau}\ln\left(\frac{1}{u}\right). \tag{23}$$

Now plug $u = \frac{d\sigma}{d\tau}$ back into the expression after the last equality of Eq. (23), we get

$$\frac{1}{u}\frac{d}{d\tau}\ln\left(\frac{1}{u}\right) = \frac{d\tau}{d\sigma}\frac{d}{d\tau}\ln\left(\frac{d\tau}{d\sigma}\right) = \frac{d}{d\sigma}\ln\left(\frac{d\tau}{d\sigma}\right). \tag{24}$$

Equation (22) therefore becomes

$$\frac{d^2x^\beta}{d\sigma^2} + \Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} = \frac{d}{d\sigma}\ln\left(\frac{d\tau}{d\sigma}\right)\frac{dx^\beta}{d\sigma}. \tag{25}$$

(b) Note that Eq. (25) is already the usual geodesic equation, except the term on the right-hand side. In other words, if we have the right-hand side term to be 0, then we get the usual geodesic equation back. The right-hand side term being 0 implies that $\ln(d\tau/d\sigma) =$ constant, which means $d\tau/d\sigma =$ constant. In other words, we have $\sigma = a\tau + b$. Therefore, if we have $\sigma = a\tau + b$, we will get the usual geodesic equation back.

6

By definition,

$$U^\alpha U_\alpha = g_{\alpha\beta} U^\alpha U^\beta. \tag{26}$$

We can choose our coordinates $\{y^{\alpha'}\}$ to be the local inertial coordinates that are centered around a point p located on the geodesic. Then in the vicinity of p, the spacetime metric is flat up to second order corrections in the size of the region. Namely,

$$g_{\alpha'\beta'} = \eta_{\alpha'\beta'} + \mathcal{O}(y^{\alpha'} - y^{\alpha'}(p))^2, \tag{27}$$

where the metric is evaluated near point p in the local inertial coordinates. In these coordinates, Eq. (26) becomes

$$U^{\alpha'} U_{\alpha'} = g_{\alpha'\beta'} U^{\alpha'} U^{\beta'} = \eta_{\alpha'\beta'} U^{\alpha'} U^{\beta'} + \mathcal{O}(y^{\alpha'} - y^{\alpha'}(p))^2. \tag{28}$$

When we evaluate $U^{\alpha'} U_{\alpha'}$ at p, then $y^{\alpha'} - y^{\alpha'}(p)$ goes to 0, so

$$U^{\alpha'} U_{\alpha'}|_p = \eta_{\alpha'\beta'} U^{\alpha'} U^{\beta'}|_p = -1, \tag{29}$$

which we know from special relativity, and Eq. (29) is exact. Note that our choice of point p is arbitrary. For any point p' along the geodesic, we can find a local inertial coordinates centered around that point, such that $U^{\alpha'} U_{\alpha'}|_{p'} = -1$. In addition, $U^{\alpha'} U_{\alpha'}$ is an invariant quantity, its value is independent of the coordinate system that we choose. Therefore we conclude that $U^\alpha U_\alpha = -1$ for a timelike geodesic in general.

7

(a) Using the given metric, we know that a radial ($d\Omega = 0$) null curve satisfies

$$ds^2 = 0 = -\left(1 + \frac{r^2}{l^2}\right) dt^2 + \frac{dr^2}{1 + r^2/l^2}. \tag{30}$$

The problem tells us that every null curve in two dimensions is a null geodesic, so Eq. (30) is already the geodesic equation. We can then solve Eq. (30) to find the trajectory of a radial light ray. Rearrange Eq. (30), we get

$$\left(\frac{dr}{dt}\right)^2 = \left(1 + \frac{r^2}{l^2}\right)^2. \tag{31}$$

For the solution of an outgoing lightray ($\frac{dr}{dt} > 0$), we get

$$\frac{dr}{dt} = 1 + \frac{r^2}{l^2} \tag{32}$$

In other words,

$$dt = \frac{dr}{1 + r^2/l^2}, \tag{33}$$

which has the solution

$$t(r) = A + l \tan^{-1}(r/l), \tag{34}$$

where A is an integration constant. Suppose the light ray was emitted from the origin at $t = 0$, then $t(r = 0) = 0$, so $A = 0$. Then our solution is

$$t(r) = l \tan^{-1}(r/l). \tag{35}$$

(b)

$$t_\infty = \lim_{r\to\infty} l \tan^{-1}(r/l) = \frac{\pi}{2} l. \tag{36}$$

Now we invert $t(r)$ and have

$$r(t) = l \tan(t/l), \tag{37}$$

so

$$r(2t_\infty) = l \tan \pi = 0. \tag{38}$$

(c) The last result of part (b) says that at time $t = 2t_\infty$, the light ray is at the origin $r = 0$. This means you see the light that was emitted at $t = 0$ has returned at where you are (the origin).

(d) For a radial geodesic condition,

$$g_{\mu\nu} U^\mu U^\nu = -1 \tag{39}$$

simplifies to

$$-\left(1 + \frac{r^2}{l^2}\right)\left(\frac{dt}{d\tau}\right)^2 + \left(1 + \frac{r^2}{l^2}\right)^{-1}\left(\frac{dr}{d\tau}\right)^2 = -1. \tag{40}$$

Rearrange the equation above, we get

$$\left(\frac{dr}{d\tau}\right)^2 = \left(1 + \frac{r^2}{l^2}\right)^2\left(\frac{dt}{d\tau}\right)^2 - \left(1 + \frac{r^2}{l^2}\right). \tag{41}$$

Now using the definition of C, we get

$$\left(\frac{dr}{d\tau}\right)^2 = C^2 - 1 - \frac{r^2}{l^2}. \tag{42}$$

At the maximum $dr/dt = 0$, we have

$$r^2 = l^2(C^2 - 1), \tag{43}$$

which is finite as long as l and C are finite. The maximum distance is

$$r_{max} = l\sqrt{C^2 - 1}. \tag{44}$$

8

A spacelike geodesic neither maximizes or minimizes the proper distance. To show this, we just need to find a spacelike geodesic between two points, and show that there exists a path connecting the two points which has a longer proper distance than that of the spacelike geodesic and find another path connecting the two points which has a shorter proper distance than that of the spacelike geodesic.

We can simply work in a local inertial coordinate system and focus on the region where we can safely ignore the deviations from flat spacetime (see Fig. [2]). We know that in flat spacetime, geodesics are straight lines, so if we pick two spacelike separated points, then the spacelike geodesic between the two points is just the line of simultaneity between them. Now we can perturb the line of simultaneity to get a new path that connects the two points but restricting the new path to be on the same plane of simultaneity that contains the two points. In other words, along the new path, $dt = 0$, so its proper length can only be larger than that of the spacelike geodesic. This shows that a spacelike geodesic does not maximize the proper length.

Now instead, we can replace some section of the spacelike geodesic with null paths so the perturbation is out of the plane of simultaneity. Since null paths have zero proper distance, the new path that we just created has shorter proper distance than that of the spacelike geodesic, so we conclude that spacelike geodesics do not minimize proper distance.

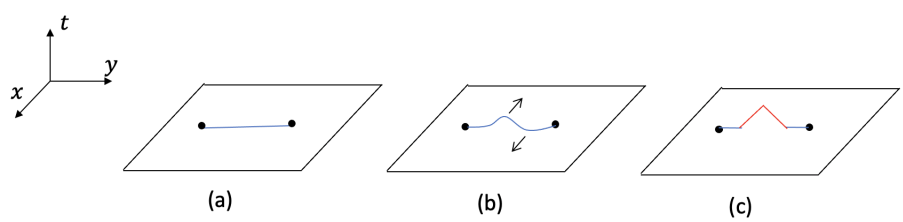

Figure 2: Select two spacelike separated points in a local inertial coordinates, where the spacetime is approximately flat. Panel (a): The geodesic between the two points is a straight line. Panel (b): Now we perturb the straight line within the plane of simultaneity. The resulting curve has a longer proper distance, so the spacelike geodesic does not maximize the proper distance. Panel (c): Instead, we can replace one segement of the straight line with null paths (the red segments), the now the resulting path has a shorter proper distance. This shows the spacelike geodesic does not minimize the proper distance.